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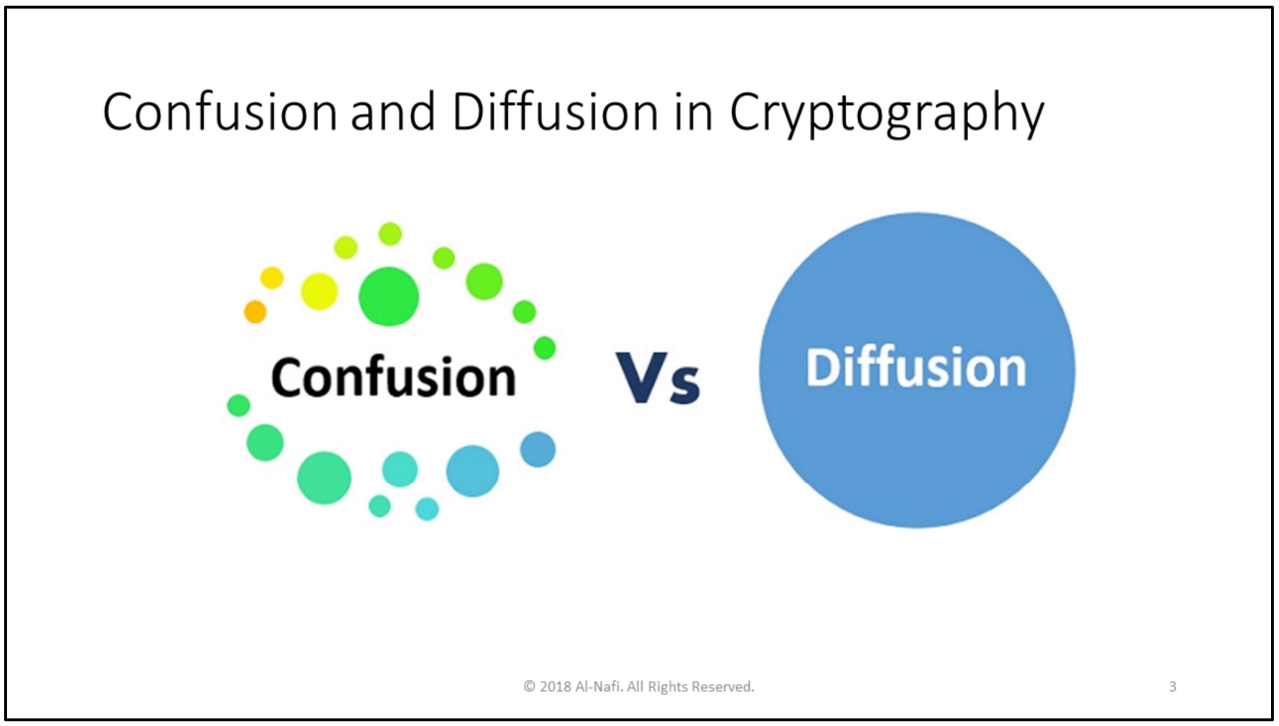
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Confusion and Diffusion in Cryptography

- Confusion means cipher text gives no clue about original text. Achieved using substitution
- Diffusion increases redundancy of plaintext by spreading across rows and columns. Achieved using transposition or permutation.

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Confusion: Provided by mixing or changing the key values used during the repeated rounds of encryption. When the key is modified for each round, it provides added complexity that the attacker would encounter.



Diffusion: Provided by mixing up the location of the plaintext throughout the ciphertext. Through transposition, the location of the first character of the plaintext may change several times during the encryption process, and this makes the cryptanalysis process much more difficult.

Avalanche effect in cryptography

- Where a change of one input or key but results in changing approx. **half** of output bits

Attack after fajr salah

1236458asdasdsa

Attack After fajr salah

Asdasdj sakdasdkla nsdajasjdcml

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Avalanche effect: An important consideration in all cryptography used to design algorithms where a minor change in either the key or the plaintext will have a significant large change in the resulting ciphertext. This is also a feature of a strong-hashing algorithm.

Example

- To check the avalanche effect in DES, let us encrypt two plaintext blocks (with the same key) that differ only in one bit and observe the differences in the number of bits in each round

Plaintext: 0000000000000000	Key: 22234512987ABB23
Ciphertext: 4789FD476E82A5F1	
Plaintext: 0000000000000001	Key: 22234512987ABB23
Ciphertext: 0A4ED5C15A63FEA3	

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Key clustering in cryptography

- Instance where two keys generate the same cipher text from the same plaintext

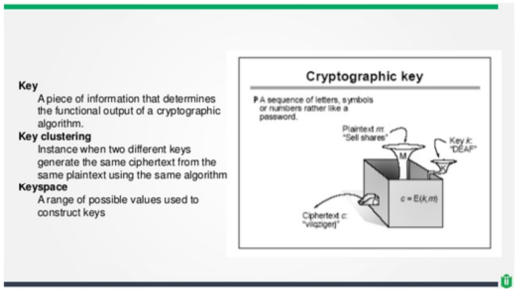
Meet me at the masjid

Key 31245678

Cipher text abscdfg

Key 311245697

Cipher text abscdfg



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Key clustering: When different encryption keys generate the same ciphertext from the same plaintext message.

Synchronous in cryptography

Each encryption or decryption request is performed immediately.

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Synchronous: Each encryption or decryption request is performed immediately.

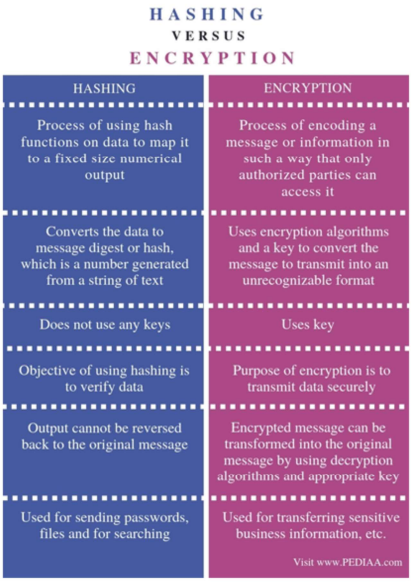
Asynchronous in cryptography

Encrypt/Decrypt requests are processed in queues. A key benefit of asynchronous cryptography is utilization of hardware devices and multiprocessor systems for cryptographic acceleration.

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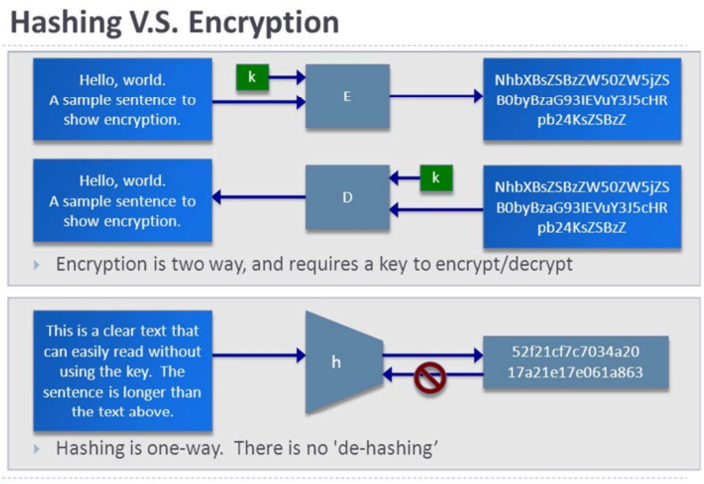
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Hash Functions vs Encryption



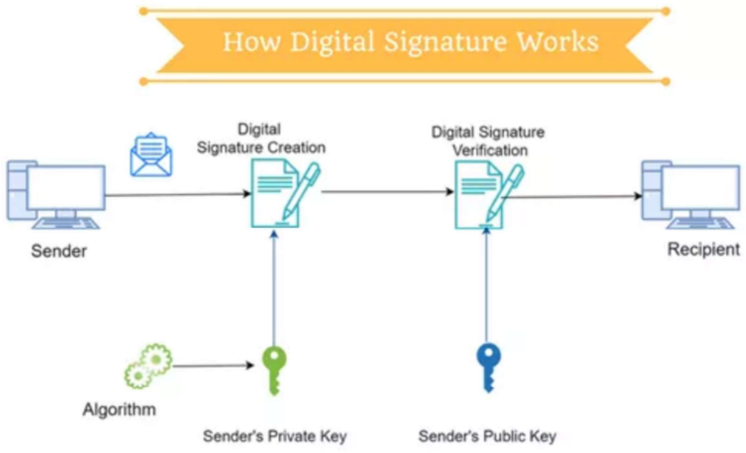
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Hashing vs Encryption



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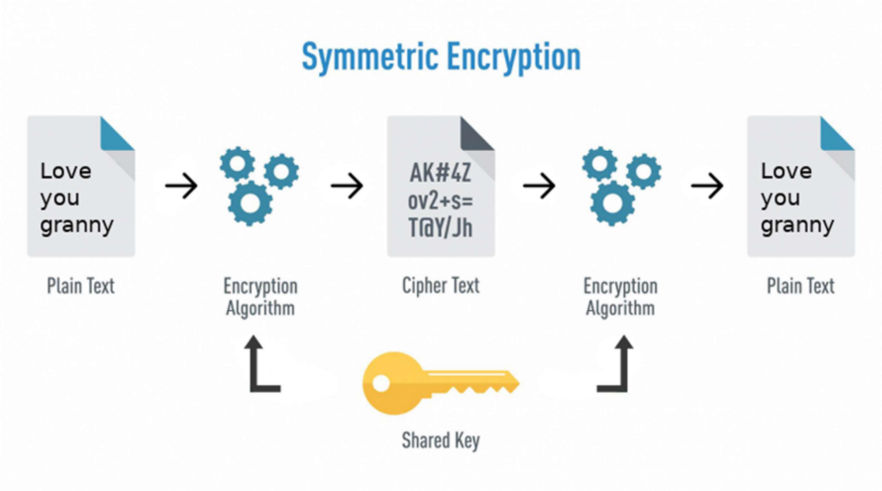
Digital Signatures



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Digital signatures: These provide authentication of a sender and integrity of a sender's message. A message is input into a hash function. Then, the hash value is encrypted using the private key of the sender. The result of these two steps yields a digital signature. The receiver can verify the digital signature by decrypting the hash value using the signer's public key, then perform the same hash computation over the message and then compare the hash values for an exact match. If the hash values are the same, then the signature is valid.

Symmetric Encryption



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Symmetric: This is a term used in cryptography to indicate that the same key is required to encrypt and decrypt. The word “symmetric” means “the same,” and we are obviously referring to the key that is required at both ends to encrypt and decrypt. Symmetric key cryptography has the fundamental problem of secure key distribution.